LECTURE – 15

BREADTH FIRST SEARCH OF A GRAPH

Simple, basic algorithm. Finds “the shortest path", in terms of number of edges, from source vertex s to every other vertex in G. Works on both directed and undirected graphs.

Meaning of “breadth first”:
This algorithm discovers all vertices at distance k from s, before discovering any vertex at distance k+1.

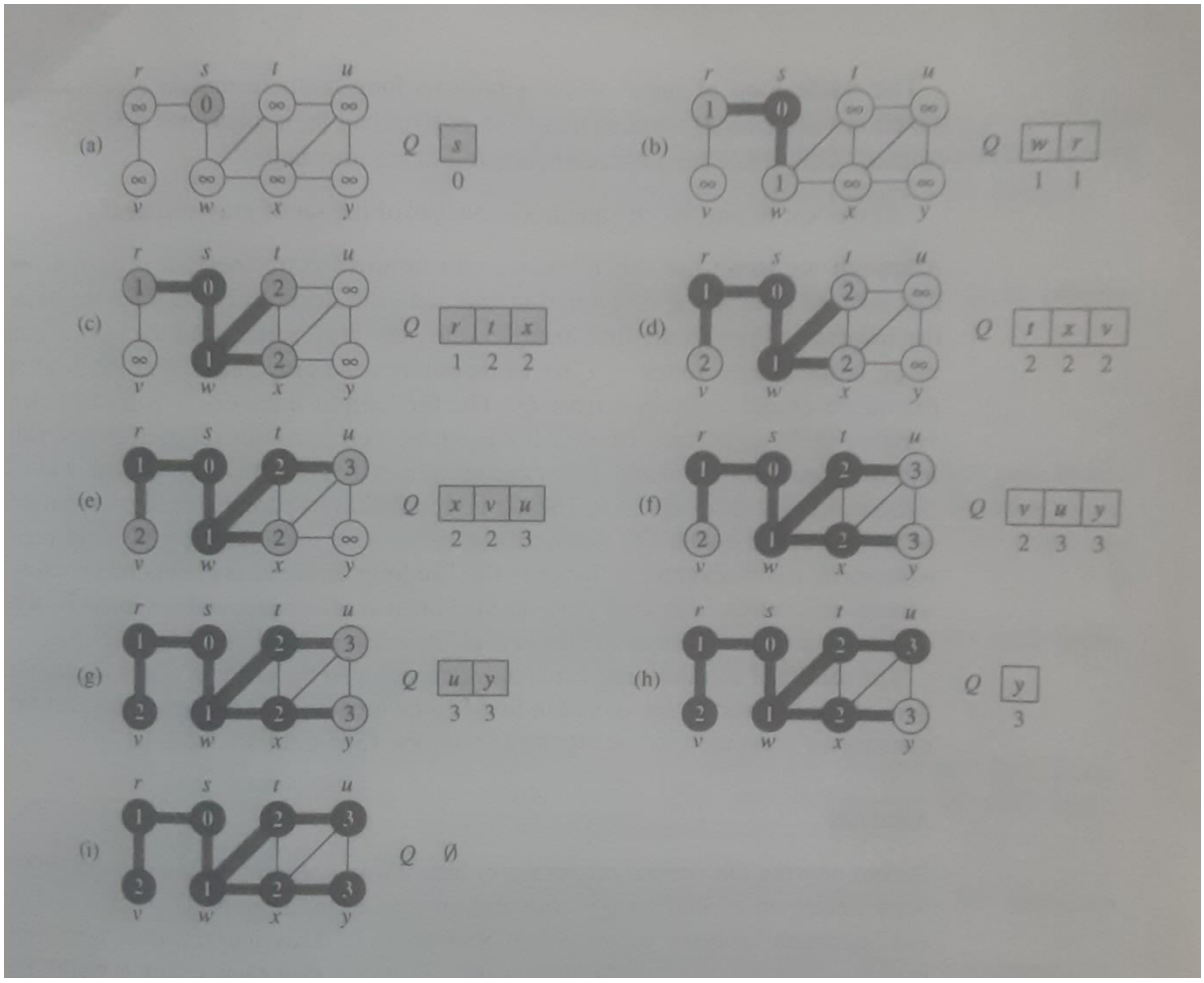
```
BFS(G,s)
  // Initialization
  for each vertex u in V[G] – {s}
    color(u) = WHITE;
    dist[u] = INF;
    pred[u] = NIL;

  color[s] = GRAY;
  dist[s] = 0;
  pred[s] = NIL;
  Q = NULL-SET;
  ENQUEUE(Q,s);

  // Now the search starts
  while Q <> NULL-SET
    // Take next node u
    u = DEQUEUE(Q);
    for each v in ADJ[u];
      if color[v] = WHITE
         color[v] = GRAY;
         dist[v] = dist[u]+1;
         pred[v] = u;
         ENQUEUE(Q,v);
    // We are done with node u
    color[u] = BLACK;
```

Loop invariant of **`while`** loop:
At the time of the **`while`** test, the queue Q consists of the set of gray vertices.

Running time analysis:

Each vertex is queued and de-queued (at most) once, each of which operations is O(1). Later, each edge is examined at most once, so the second part of the algorithm is O(E). Hence the total running time is O(V+E).

Example:



Provable properties of DFS

Notation: The **shortest path distance** $\delta(s,v)$ from source vertex s to any other vertex v is the minimum number of edges in any path from s to v.

1. For any edge (u,v) in E, $\delta(s,v) \leq \delta(s,u)+1$.

2. After BFS is run, every vertex v in V that is reachable from s is discovered, and dist[v] = $\delta(s,v)$.

3. The predecessor relationship discovered by BFS defines a breadth first tree of graph G with source vertex s.

Function **`print-path(G,s,v)`** → self- study.